

FROM SUPPRESSION TO REPAIR: MITIGATING HALLUCINATION IN LARGE VISION-LANGUAGE MODELS VIA LOCALIZED DISTRIBUTION ALIGNMENT

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ABSTRACT

Object hallucination in large vision-language models (LVLMs) remains a major obstacle to trustworthy generation. An intuitive mitigation strategy is to suppress the components in the representations that are associated with hallucinations. However, since these components may also contain useful visual-semantic information, directly suppressing them can impair the model’s multimodal capabilities. We therefore propose **ResOT**, a training-free inference-time method that replaces suppression with localized distribution repair. Specifically, ResOT confines repair to a low-dimensional residual subspace that captures hallucination-associated structures not explained by the principal subspace of faithful representations. In this subspace, a closed-form Gaussian optimal transport (OT) map aligns the hallucinated distribution with the faithful one while minimizing displacement. At inference, ResOT adjusts the repair strength for each token state based on its hallucination tendency. Across multiple benchmarks and three representative LVLMs, ResOT achieves the best overall performance in hallucination mitigation and captioning among the compared methods. It also improves multimodal performance over the corresponding vanilla LVLMs while incurring low inference overhead.

1 INTRODUCTION

Large vision-language models (LVLMs) have achieved strong performance in image captioning, visual question answering, and visual reasoning (Alayrac et al., 2022; Bai et al., 2023; Chen et al., 2024b). With their powerful multimodal alignment capabilities, LVLMs can generate fluent and informative responses given an input image. However, these responses are not always faithful to the image content. LVLMs may mention objects that do not appear in the image, a failure known as object hallucination (Bai et al., 2024; Li et al., 2023; Rohrbach et al., 2018). Such errors undermine the reliability of LVLMs and motivate research on hallucination mitigation.

A straightforward solution is to fine-tune LVLMs using additional data with hallucination annotations, but this alters the parameters of the pretrained models and incurs substantial training costs (Jiang et al., 2024; Liu et al., 2024a). To avoid retraining, recent work has explored inference-time interventions that mitigate hallucinations by directly modifying hidden representations (Liu et al., 2025; Lin et al., 2026). These methods commonly identify directions or subspaces associated with hallucinations and suppress the corresponding components (Yang et al., 2025a; Zhu et al., 2026; Dastmalchi et al., 2026). However, these components are not necessarily specific to hallucination; they can also encode useful visual-semantic information (Hua et al., 2025). Consequently, directly suppressing them may harm the model’s multimodal capabilities. This limitation has prompted us to shift from suppression to repair: correcting hallucination-associated deviations while preserving the visual-semantic structure shared with faithful representations.

Fig. 1 illustrates how suppression can be destructive. In general, a hallucinated response is not entirely incorrect. It may accurately describe the scene in the image, including the attributes and relations of existing objects, but introduce nonexistent objects. Therefore, the representations may retain visual-semantic structures shared with faithful representations, resulting in the partial overlap shown in Fig. 1(a). However, direct suppression overlooks this overlap. As shown in Fig. 1(b), suppressing the hallucination-associated direction also shifts faithful representations in the overlap-

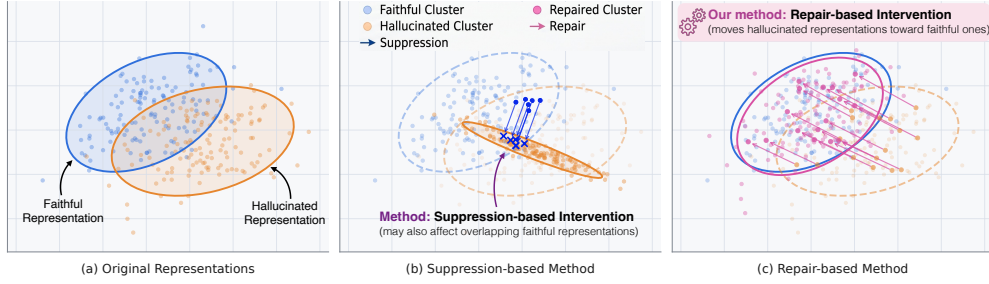


Figure 1: Comparison of suppression and repair. (a) Faithful and hallucinated representations form two distinct but partially overlapping clusters. (b) Suppression maps the hallucinated representations into a narrow band. The blue arrows show how faithful representations in the overlapping region are moved by the same operation. (c) Repair-based intervention moves hallucinated representations toward the faithful distribution, while preserving the structure of the faithful cluster.

ping region, potentially discarding useful visual-semantic information. By contrast, the repair-based intervention shown in Fig. 1(c) moves hallucinated representations toward the faithful distribution while preserving the shared visual-semantic structure.

In this paper, we propose **ResOT**, a lightweight, training-free method that repairs representations during inference. Our method focuses on two questions: *where to repair* and *how to repair*. To answer the first question, ResOT identifies a low-dimensional residual subspace for repair. This subspace specifically captures deviations associated with hallucinations while excluding the dominant structures of faithful representations. Within this subspace, ResOT models the hallucinated and faithful distributions as Gaussians and derives a closed-form optimal transport (OT) map. This map defines a full repair target that moves the hallucinated distribution to the faithful one with minimal displacement. However, not every token state requires the full OT repair. At inference, a likelihood-based gate controls the repair strength for each token state. As a result, states that are more likely to be hallucinated receive stronger repair. All updates are confined to the residual subspace, leaving the orthogonal components unchanged.

Experiments conducted on three representative LVLMs demonstrate that ResOT reduces hallucinations without sacrificing caption quality or multimodal performance. Across all three LVLMs, ResOT achieved the highest POPE F1 score and the lowest CHAIR_s and CHAIR_i among the compared methods (Li et al., 2023; Rohrbach et al., 2018). It also improved the BLEU score relative to the corresponding base models, and open-ended evaluations showed that the generated responses remained detailed (Papineni et al., 2002). Furthermore, ResOT increased the MME score by 4.77%, indicating that the reduction in hallucinations did not come at the expense of overall multimodal capability (Fu et al., 2023). These improvements require only 8.15 MiB of peak GPU memory while retaining 81.24% of the base model’s decoding speed. Ablation studies show that these performance gains are attributable to two key design choices: the residual subspace provides a more targeted intervention space, and within this space, transport-based repair methods consistently outperform direct suppression methods.

2 RELATED WORK

2.1 HALLUCINATION IN LARGE VISION-LANGUAGE MODELS

Object hallucination may originate from various stages of LVLM processing. Weak visual perception, cross-modal misalignment, or strong language prior can lead the model to generate content inconsistent with the image (Gu et al., 2025; Jain et al., 2024; Xing et al., 2024; Liu et al., 2024b). Training-based methods address these errors through supervised fine-tuning (Liu et al., 2024a), contrastive learning (Jiang et al., 2024), RLHF (Sun et al., 2024), or DPO (Yang et al., 2025b; Yu et al., 2024). However, these methods require additional supervision and parameter optimization, resulting in high training costs. Training-free methods, on the other hand, intervene during or after generation. Specifically, decoding-time approaches adjust token selections using visual contrast (Leng et al., 2024; Chen et al., 2024a), intermediate-layer predictions (Wang et al., 2025; Chuang et al.,

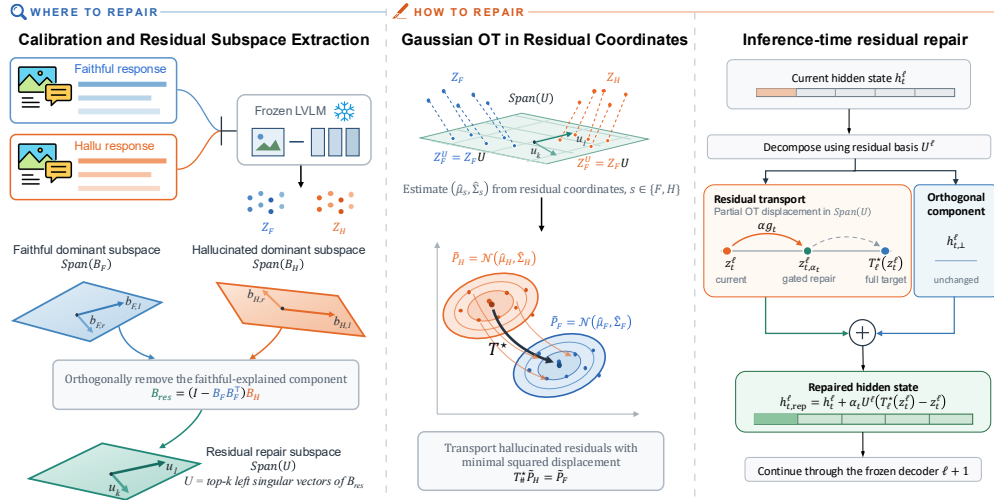


Figure 2: Overview of ResOT. Paired calibration representations identify a hallucination-dominant residual subspace after filtering out dominant faithful structure. A closed-form Gaussian OT map defines the minimum-displacement full repair target in this subspace. At inference, likelihood gating applies an adaptive fraction of the OT displacement, leaving the orthogonal component unchanged.

2024), or attention signals (Zhou et al., 2025; Huang et al., 2024). Post-hoc approaches verify and revise unsupported content after generation (Zhou et al., 2024; Yin et al., 2024). These methods avoid retraining but may increase inference costs due to additional forward passes, complex decoding, or external verification.

ResOT is mostly related to the representation-level intervention, which directly modifies hidden states. Existing methods operate by steer latent activations (Liu et al., 2025; Li et al., 2025), intervene on attention heads (Chen et al., 2025), or suppress hallucination-associated components through static or adaptive subspace editing (Yang et al., 2025a; Zhu et al., 2026; Lin et al., 2026; Dastmalchi et al., 2026). However, a component associated with hallucination is not necessarily specific to it. The same component may also encode useful visual-semantic information. ResOT instead isolates hallucinated dominant residuals orthogonal to the dominant faithful structure and repairs their distribution through optimal transport. Thus, it differs from suppression in both where it intervenes and how it modifies the representation.

2.2 OPTIMAL TRANSPORT AND DISTRIBUTIONAL REPRESENTATION ALIGNMENT

Optimal transport (OT) aligns distributions by minimizing transport cost under the geometry of the underlying space (Villani, 2009; Peyré & Cuturi, 2019). It is widely used for feature alignment and domain adaptation (Courty et al., 2017; Damodaran et al., 2018). Empirical OT computes a transport plan across source and target samples, which can be costly at scale. Sinkhorn iterations improve scalability but still require iterative optimization (Cuturi, 2013). For Gaussian distributions under quadratic cost, the optimal map has a closed-form affine solution determined by their means and covariances (Peyré & Cuturi, 2019). ResOT exploits this structure within a low-dimensional residual subspace, using the transport map as a full repair target and applying a likelihood-gated fraction of its displacement at inference without iterative OT during generation.

3 METHOD

This section presents ResOT, including its calibration and inference procedures. We then analyze the advantages of residual-subspace construction and Gaussian OT repair over projection-based suppression. Figure 2 provides an overview.

3.1 LVLM GENERATION AND CALIBRATION REPRESENTATIONS

LVLMs typically comprise a vision encoder, a vision-language connector, and a language decoder (Radford et al., 2021; Liu et al., 2023). The vision encoder extracts features from the visual input $\mathcal{V}$, and the connector adapts them for the language decoder. Given the projected visual features and a text prompt x , the language decoder generates a response autoregressively. At decoding step t , let $h_t^\ell \in \mathbb{R}^d$ denote the hidden state at decoder layer ℓ . This state is propagated through subsequent decoder layers to predict the next token y_t .

Following prior projection-based methods (Yang et al., 2025a; Zhu et al., 2026), each calibration image is paired with a faithful and a hallucinated description. For each selected layer ℓ , we pass these image-description pairs through the frozen LVLM and collect their final-position hidden states. Stacking these states yields $Z_F^\ell, Z_H^\ell \in \mathbb{R}^{N \times d}$. ResOT uses them to estimate a layer-specific residual subspace and Gaussian OT map. At inference, a gated update derived from this map modifies h_t^ℓ before propagation to subsequent layers. Model parameters remain frozen throughout.

3.2 RESIDUAL-SUBSPACE REPRESENTATION REPAIR

3.2.1 LOCATING THE REPAIR SUBSPACE

ResOT confines repair to hallucination-dominant directions that are not explained by the dominant faithful subspace. Given $Z_F, Z_H \in \mathbb{R}^{N \times d}$, we omit the layer index ℓ for clarity. Their dominant representation subspaces are obtained from

$$Z_F = L_F S_F R_F^\top, \quad Z_H = L_H S_H R_H^\top. \quad (1)$$

With the right singular vectors ordered by corresponding singular value, we retain the top- c columns of R_F and R_H to form $B_F = [b_{F,1}, b_{F,2}, \dots, b_{F,r}] \in \mathbb{R}^{d \times r}$ and $B_H = [b_{H,1}, b_{H,2}, \dots, b_{H,r}] \in \mathbb{R}^{d \times r}$, where B_F and B_H are orthonormal bases for the dominant faithful and hallucinated subspaces. The component of B_H explained by the faithful subspace is $B_F B_F^\top B_H$. Removing this component gives

$$B_{\text{res}} = B_H - B_F (B_F^\top B_H) = (I_d - B_F B_F^\top) B_H. \quad (2)$$

We compute the SVD $B_{\text{res}} = \bar{L} \bar{S} \bar{R}^\top$ and let $q = \text{rank}(B_{\text{res}})$. For $k \leq q$, the residual basis is $U = [\bar{L}[:, 1 : k], 1 : k] \in \mathbb{R}^{d \times k}$. Its columns are orthonormal and satisfy $B_F^\top U = 0$. The corresponding singular values measure how strongly the dominant hallucinated directions extend beyond the faithful subspace. All subsequent interventions are confined to $\text{span}(U)$.

3.2.2 GAUSSIAN OPTIMAL TRANSPORT IN THE RESIDUAL SUBSPACE

Within the residual subspace, Gaussian OT defines a full repair target that aligns both the mean and covariance without removing residual directions. We project the calibration representations onto $\text{span}(U)$: $Z_F^U = Z_F U$ and $Z_H^U = Z_H U$, where each row is a k -dimensional residual coordinate. Let $(\hat{\mu}_F, \hat{\Sigma}_F)$ and $(\hat{\mu}_H, \hat{\Sigma}_H)$ denote their empirical means and covariances. Finite calibration data may produce noisy or ill-conditioned covariance estimates. We therefore apply covariance shrinkage (Ledoit & Wolf, 2004) before computing the affine OT map and the Gaussian log-likelihoods used by the inference-time gate:

$$\hat{\Sigma}_s \leftarrow (1 - \rho) \hat{\Sigma}_s + \rho \frac{\text{tr}(\hat{\Sigma}_s)}{k} I_k + \epsilon I_k, \quad s \in \{F, H\}. \quad (3)$$

Here, $\rho \in [0, 1]$ is the shrinkage coefficient and $\epsilon > 0$ ensures positive definiteness. We then approximate the residual distributions as

$$\tilde{P}_F = \mathcal{N}(\hat{\mu}_F, \hat{\Sigma}_F), \quad \tilde{P}_H = \mathcal{N}(\hat{\mu}_H, \hat{\Sigma}_H). \quad (4)$$

We seek a transport map that aligns the hallucinated approximation with the faithful one while minimizing expected squared displacement:

$$T^* = \arg \min_{T \in \mathcal{T}} \mathbb{E}_{z \sim \tilde{P}_H} \|T(z) - z\|_2^2 \quad \text{s.t.} \quad T_\# \tilde{P}_H = \tilde{P}_F. \quad (5)$$

The constraint requires the transported Gaussian approximation to match the faithful Gaussian approximation. For non-degenerate Gaussian distributions, the optimal map has the closed-form affine solution (Brenier, 1991; Peyré & Cuturi, 2019):

$$T^*(z) = \hat{\mu}_F + A(z - \hat{\mu}_H), \quad (6)$$

where $A = \hat{\Sigma}_H^{-\frac{1}{2}} \left(\hat{\Sigma}_H^{\frac{1}{2}} \hat{\Sigma}_F \hat{\Sigma}_H^{\frac{1}{2}} \right)^{\frac{1}{2}} \hat{\Sigma}_H^{-\frac{1}{2}}$. Thus, T^* exactly transports $\tilde{P}_H$ to $\tilde{P}_F$, i.e., $T^*_{\#} \tilde{P}_H = \tilde{P}_F$, matching both the mean and covariance of the faithful Gaussian approximation. Among maps satisfying this constraint, T^* minimizes expected squared displacement and therefore defines the minimum-displacement full repair target in the residual subspace.

3.2.3 REPAIR AT INFERENCE

Each selected layer ℓ has a residual basis U^ℓ and an OT map T_ℓ^* , both estimated offline. At decoding step t , the residual coordinate and orthogonal component are

$$z_t^\ell = (U^\ell)^\top h_t^\ell, \quad h_{t,\perp}^\ell = (I_d - U^\ell (U^\ell)^\top) h_t^\ell. \quad (7)$$

The full repair target at this layer is $\tilde{z}_t^\ell = T_\ell^*(z_t^\ell)$. However, the required repair strength varies across decoding states. ResOT therefore computes a lightweight gate from the pre-repair state at the first selected layer ℓ_a and shares it across all selected layers:

$$s_t = \frac{1}{k} \left[\log \mathcal{N}(z_t^{\ell_a}; \hat{\mu}_H^{\ell_a}, \hat{\Sigma}_H^{\ell_a}) - \log \mathcal{N}(z_t^{\ell_a}; \hat{\mu}_F^{\ell_a}, \hat{\Sigma}_F^{\ell_a}) \right], \quad g_t = \sigma(s_t). \quad (8)$$

A larger g_t indicates stronger relative evidence for the hallucinated approximation, so a larger fraction of the OT displacement is applied. We define the effective repair strength as $\alpha_t = \alpha g_t$, where $\alpha \in [0, 1]$ controls the maximum displacement. We then interpolate between the original and transported residual coordinates:

$$z_{t,\alpha_t}^\ell = (1 - \alpha_t) z_t^\ell + \alpha_t \tilde{z}_t^\ell, \quad (9)$$

Combining this coordinate with the unchanged orthogonal component gives

$$h_{t,\text{rep}}^\ell = h_t^\ell + \alpha_t U^\ell (T_\ell^*(z_t^\ell) - z_t^\ell). \quad (10)$$

At inference, we cap the OT correction norm at a threshold estimated for each layer from hallucinated calibration states, preserving its direction and preventing unusually large updates. Thus, ResOT adaptively scales the OT displacement for each token within the residual subspace. The orthogonal component remains unchanged, and the repaired state propagates through subsequent frozen decoder layers.

3.2.4 THEORETICAL ANALYSIS

The following theory justify ResOT's residual subspace and Gaussian OT repair. First, we show that the residual basis maximizes alignment with the dominant hallucinated subspace while remaining orthogonal to the dominant faithful subspace. Second, we show that hard projection-based suppression incurs an unavoidable mismatch whenever it removes faithful energy. Finally, we establish that Gaussian OT is the minimum-displacement affine map among those matching the faithful mean and covariance, even for a non-Gaussian source distribution. All proofs are provided in Appendix A.

Proposition 1. Let $B_{\text{res}} = (I_d - B_F B_F^\top) B_H$, and $U \in \mathbb{R}^{d \times k}$ contain its top- k left singular vectors, where $k \leq \text{rank}(B_{\text{res}})$. For any orthonormal basis $V \in \mathbb{R}^{d \times k}$ satisfying $B_F^\top V = 0$,

$$\|V^\top B_H\|_F^2 \leq \|U^\top B_H\|_F^2 = \sum_{i=1}^k \sigma_i^2(B_{\text{res}}).$$

Thus, among all k -dimensional subspaces orthogonal to the dominant faithful subspace, $\text{span}(U)$ has the greatest squared alignment with the dominant hallucinated basis.

Proposition 2. Let $\tilde{P}_H, \tilde{P}_F \in \mathcal{P}_2(\mathbb{R}^k)$, and let μ_F and Σ_F be the mean and covariance of $\tilde{P}_F$. For $T_\Pi(z) = \Pi z$, where $\Pi = \Pi^\top = \Pi^2$ is a rank-reducing orthogonal projection,

$$W_2^2((T_\Pi)_\# \tilde{P}_H, \tilde{P}_F) = W_2^2((T_\Pi)_\# \tilde{P}_H, (T_\Pi)_\# \tilde{P}_F) + \|(I_k - \Pi)\mu_F\|_2^2 + \text{tr}((I_k - \Pi)\Sigma_F).$$

The first term is the Wasserstein mismatch between the projected distributions. The last two terms equal the faithful energy removed by suppression. Therefore, whenever the faithful distribution has nonzero energy in the removed subspace, suppression leaves a strictly positive mismatch to the original faithful distribution, even if the projected distributions match exactly.

Proposition 3. Let $Z \in \mathbb{R}^k$ be any random vector with finite second moment, mean μ_H and covariance $\Sigma_H \in \mathbb{S}_{++}^k$. Given a target $\mu_F \in \mathbb{R}^k$ and $\Sigma_F \in \mathbb{S}_{++}^k$, define

$$\mathcal{A}_{H \rightarrow F} = \{T_A(z) = \mu_F + A(z - \mu_H) \mid A\Sigma_H A^\top = \Sigma_F\}.$$

Then the Gaussian OT map T^* is the unique minimizer:

$$T^* = \arg \min_{T_A \in \mathcal{A}_{H \rightarrow F}} \mathbb{E} [\|T_A(Z) - Z\|_2^2].$$

When $Z \sim \mathcal{N}(\mu_H, \Sigma_H)$ and the target is $\mathcal{N}(\mu_F, \Sigma_F)$, T^* is the exact quadratic-cost OT map. For arbitrary Z , it still uniquely minimizes displacement among affine maps matching the target mean and covariance, but exact matching of a full target distribution is not generally guaranteed.

4 EXPERIMENTS

We organize our evaluation around three questions. **RQ1:** Does ResOT reduce object hallucination while preserving caption quality? **RQ2:** Does it retain general multimodal capability and open-ended generation quality? **RQ3:** Why does ResOT outperform direct suppression?

4.1 EXPERIMENTAL SETUP

Baselines. We compare ResOT against vanilla LVLMS and two categories of training-free mitigation methods. The vanilla baselines are LLaVA-1.5 (Liu et al., 2023), mPLUG-Owl2 (Ye et al., 2024), and InstructBLIP (Dai et al., 2023), all equipped with 7B language decoders and evaluated using greedy decoding. Decoding-time baselines include OPERA (Huang et al., 2024), VCD (Leng et al., 2024), HALC (Chen et al., 2024a), and LVLMS-Saliency (Zhang et al., 2026). Representation-level baselines include ICT (Chen et al., 2025), VTI (Liu et al., 2025), VISTA (Li et al., 2025), Nullu (Yang et al., 2025a), HulluEdit (Lin et al., 2026), and REVIS (Wu et al., 2026), which provide the most direct comparison to ResOT.

Benchmarks and Metrics. We evaluate ResOT along three dimensions: object hallucination, generation quality, and general multimodal capability. POPE (Li et al., 2023) measures discriminative object hallucination through yes/no object-existence questions. We report accuracy, precision, and F1 score. CHAIR (Rohrbach et al., 2018) evaluates hallucinated objects in open-ended captions using CHAIR_s (C_s) and CHAIR_i (C_i), while BLEU (Papineni et al., 2002) evaluates caption quality. MME (Fu et al., 2023) evaluates multimodal perception and cognition. We further use GPT-4o to assess the visual accuracy and detailedness of open-ended descriptions on sampled COCO images (Lin et al., 2014). Broader evaluations on MM-Vet, MMHal-Bench, and AMBER (Yu et al., 2023; Sun et al., 2024; Wang et al., 2023) are provided in Appendix C.1.

4.2 OBJECT HALLUCINATION AND CAPTION QUALITY

We evaluate ResOT in two complementary settings. POPE measures discriminative object hallucination, while CHAIR and BLEU assess hallucination and caption quality in open-ended generation.

Results on POPE. Table 1 reports results averaged over the random, popular, and adversarial splits. ResOT achieves the strongest overall performance across all three LVLMS backbones. It achieves the highest accuracy and F1 score in every case, together with the highest precision on mPLUG-Owl2 and InstructBLIP. These improvements show that residual subspace repair effectively reduces object hallucination while maintaining accurate object recognition.

Results on CHAIR. Table 2 reports the COCO captioning results. Across all three LVLMS backbones, ResOT achieves the lowest C_s , C_i and the highest BLEU. This improvement demonstrates the central advantage of repair: ResOT improves caption faithfulness without sacrificing the visual-semantic information needed for high-quality generation.

Table 1: Average POPE results across the random, popular and adversarial splits. Higher values indicate better performance. The best results are highlighted in bold.

Method	LLaVA-1.5			mPLUG-Owl2			InstructBLIP		
	Acc. $\uparrow$	Prec. $\uparrow$	F1 $\uparrow$	Acc. $\uparrow$	Prec. $\uparrow$	F1 $\uparrow$	Acc. $\uparrow$	Prec. $\uparrow$	F1 $\uparrow$
Vanilla	84.20	82.30	84.67	78.23	72.07	81.07	84.17	83.07	84.60
OPERA	84.67	83.37	85.13	79.77	74.37	82.03	85.00	83.93	85.43
VCD	81.13	77.83	82.30	76.73	70.83	79.77	75.97	78.20	81.63
HALC	84.83	82.80	85.53	78.30	72.23	81.13	84.60	83.37	85.13
VISTA	84.23	80.03	85.47	76.00	69.33	79.70	85.20	84.87	85.50
ICT	85.43	83.10	86.07	79.07	73.37	81.57	55.50	53.00	69.20
VTI	85.21	81.73	86.23	80.09	79.75	81.96	83.70	81.00	84.63
Nullu	85.53	86.07	85.70	77.87	71.60	80.87	84.63	83.20	85.20
Revis	85.19	82.15	86.09	78.24	72.10	81.12	84.82	83.78	85.30
HulluEdit	85.63	82.76	86.46	80.87	72.36	81.27	83.08	80.25	84.14
LVLMS-Saliency	85.12	81.96	86.05	78.01	71.78	80.98	84.46	83.09	85.03
ResOT	86.13	84.43	86.67	82.50	81.57	83.60	86.63	86.93	86.40

Table 2: Captioning results on COCO evaluated by CHAIR and BLEU. Lower C_s and C_i indicate fewer hallucinated objects, while higher BLEU indicates better caption quality. The maximum generation length is set to 512 tokens. The best results are highlighted in bold.

Method	LLaVA-1.5			mPLUG-Owl2			InstructBLIP		
	$C_s \downarrow$	$C_i \downarrow$	BLEU $\uparrow$	$C_s \downarrow$	$C_i \downarrow$	BLEU $\uparrow$	$C_s \downarrow$	$C_i \downarrow$	BLEU $\uparrow$
Vanilla	0.514	0.1451	0.1739	0.568	0.1639	0.1559	0.462	0.1310	0.1740
OPERA	0.454	0.1317	0.1769	0.488	0.1503	0.1769	0.398	0.1242	0.1524
VCD	0.484	0.1437	0.1743	0.580	0.1808	0.1689	0.566	0.1808	0.1698
HALC	0.380	0.1270	0.1870	0.500	0.1432	0.1762	0.620	0.1770	0.1580
VISTA	0.436	0.1336	0.1619	0.528	0.1638	0.1426	0.434	0.1345	0.1508
ICT	0.498	0.1439	0.1797	0.716	0.2037	0.1585	0.482	0.1349	0.1196
VTI	0.322	0.1195	0.2017	0.314	0.1194	0.2087	0.428	0.1236	0.1379
Nullu	0.394	0.1150	0.1581	0.412	0.1395	0.1830	0.440	0.1429	0.1760
Revis	0.430	0.1361	0.1740	0.460	0.1489	0.1711	0.440	0.1211	0.1744
HulluEdit	0.404	0.1287	0.1669	0.362	0.1324	0.1515	0.442	0.1203	0.1633
LVLMS-Saliency	0.348	0.1132	0.2091	0.330	0.1188	0.1798	0.436	0.1288	0.1757
ResOT	0.292	0.0953	0.2334	0.258	0.0898	0.2504	0.378	0.1076	0.1770

4.3 GENERAL MULTIMODAL CAPABILITY AND OPEN-ENDED GENERATION

To answer RQ2, we evaluate whether ResOT preserves general multimodal capability and improves visual faithfulness without sacrificing detail in open-ended generation.

Results on MME. Figure 3 reports the MME results. Compared with the vanilla models, ResOT improves the total, perception, and cognition scores across all three backbones. It achieves the highest total and perception scores on LLaVA-1.5 and mPLUG-Owl2, together with the best or tied-best results in most object related subcategories. These improvements show that ResOT improves visual faithfulness without compromising broader multimodal capability.

GPT-4o Evaluation on COCO. Following prior protocols (Wang et al., 2025; Yang et al., 2025a), we evaluate open-ended descriptions on the 500 COCO validation images selected by OPERA. We use the same evaluation prompt as prior work, repeat the scoring five times, and report the mean scores. We compare ResOT with the vanilla backbone and Nullu, a representative suppression-based method. As shown in Table 3, ResOT achieves higher accuracy than both baselines across all three backbones and higher detailedness in five of the six comparisons. These results demonstrate improved visual faithfulness while retaining informative content.

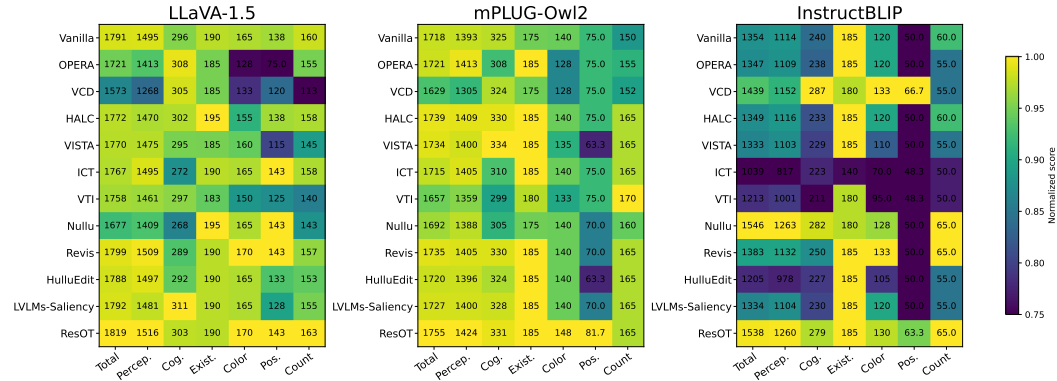


Figure 3: MME results across three LVM backbones. Each cell reports the raw score, while color intensity indicates the normalized score within each metric. We report the total, perception and cognition scores, together with four perception subcategories: existence, color, position and count.

Table 3: GPT-4o evaluation on COCO image descriptions. Each entry reports the baseline/ResOT scores for visual accuracy (Acc.) and detailedness (Det.); higher values indicate better performance.

Backbone	Vanilla vs. ResOT		Nullu vs. ResOT	
	Acc.	Det.	Acc.	Det.
LLaVA-1.5	5.72 / 6.77	6.09 / 5.81	6.22 / 6.32	5.76 / 5.82
mPLUG-Owl2	5.65 / 6.52	5.90 / 5.94	6.38 / 6.88	6.22 / 6.24
InstructBLIP	5.68 / 6.38	5.82 / 5.94	5.36 / 6.48	5.72 / 6.00

4.4 REPRESENTATION ANALYSIS AND ABLATION STUDY

To answer RQ3, we examine whether the advantage of ResOT arises from both its intervention space and its repair operation. We visualize how the residual repair subspace exposes hallucination-related mismatch and evaluate calibration efficiency and robustness. Additional analyses of Gaussian fit and hyperparameter sensitivity are provided in Appendix D.

Representation Distribution Visualization. For a held-out analysis, we use 4000 calibration pairs to estimate the subspaces and repair parameters, while the remaining 1000 pairs for visualization. Figure 4 shows that faithful and hallucinated representations are highly mixed in the Nullu-style subspace. After removing components explained by the dominant faithful subspace, the residual repair subspace reveals a clearer distributional shift. Nevertheless, the two distributions still overlap because the residual subspace concentrates their dominant mismatch rather than separating them completely, and its orthogonality to the dominant faithful basis does not make all faithful residual coordinates vanish. Direct suppression would remove components also present in faithful representations along with hallucination-related deviations. ResOT instead uses a likelihood-gated OT update to move hallucinated representations toward the faithful distribution without projecting away the residual subspace. The visualization thus shows both how ResOT localizes the mismatch and why repair is preferable to suppression.

Ablation Study. We ablate the intervention space, repair operation, and adaptive gate. For subspace construction, we compare random, Nullu-style, and residual subspaces using Gaussian OT without gating. For the repair operation, we fix the residual subspace and compare direct suppression, mean-only alignment, and Gaussian OT, all without gating. Finally, we compare Gaussian OT with and without the gate

Table 4: Ablation results on subspace construction and intervention operations.

Variant	$C_s \downarrow$	BLEU $\uparrow$	MME $\uparrow$
RandSub + OT	0.516	0.1748	1787.17
NulluSub + OT	0.424	0.1681	1764.82
ResSub + Sup	0.378	0.1903	1784.96
ResSub + Mean	0.320	0.2140	1798.35
ResSub + OT w/o Gate	0.292	0.2268	1814.17
ResSub + OT w/ Gate	0.292	0.2334	1819.02



Figure 4: Projection-space visualization of faithful, hallucinated and repaired representations at layer 31. The three panels show the Nullu suppressed space, the ResOT residual subspace before repair, and the ResOT residual subspace after Gaussian OT repair.

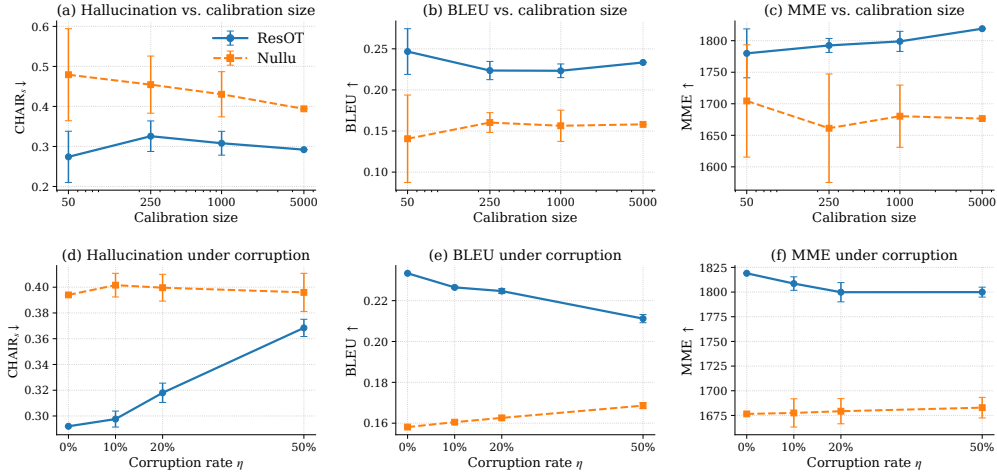


Figure 5: Effect of calibration size and corruption. We compare ResOT and Nullu under different calibration sizes and corruption rates. Error bars denote the standard deviation across random trials.

in the residual subspace. Table 4 validates all three components. With Gaussian OT fixed, the residual subspace achieves the best hallucination, captioning, and MME results among the three subspace constructions. Within the residual subspace, Gaussian OT consistently outperforms suppression and mean-only alignment, confirming the benefit of covariance alignment beyond mean correction. The adaptive gate further increases BLEU and MME while retaining the same C_s . These results show that the residual subspace localizes where to repair, Gaussian OT determines how to repair, and the gate limits unnecessary changes.

Calibration Set Analysis. We re-estimate ResOT using nested calibration subsets of different sizes. To evaluate robustness to imperfect calibration, we corrupt a fraction η of the calibration pairs through image–text mismatches and faithful/hallucinated label swaps. Figure 5 shows that ResOT establishes a clear advantage with only 50 calibration pairs, achieving lower C_s and higher BLEU and MME than Nullu. Additional calibration data primarily improves estimation stability. ResOT also retains its advantage on all three metrics under corruption rates up to 50%, with its MME score remaining close to 1800. These results demonstrate that ResOT is highly efficient in calibration and robust to imperfect calibration data.

5 CONCLUSION

ResOT establishes a precise intervention principle for hallucination mitigation: operate only within a localized residual space and only to the degree indicated by each token’s hallucination tendency.

Gaussian optimal transport realizes this principle as a minimum-displacement repair. And theoretical results characterizing the optimality of both localization and transport. Consistent empirical gains further establish localized repair as an effective alternative to direction suppression.

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A PROOF

A.1 PROOF OF PROPOSITION 1

Since B_F has orthonormal columns,

$$B_F^\top B_{\text{res}} = B_F^\top (I_d - B_F B_F^\top) B_H = 0.$$

Hence, the left singular vectors of B_{res} associated with nonzero singular values are orthogonal to $\text{Span}(B_F)$. Because $k \leq \text{rank}(B_{\text{res}})$, $B_F^\top U = 0$.

For any feasible V , $B_F^\top V = 0$ implies $V^\top B_F B_F^\top = 0$. Thus,

$$V^\top B_H = V^\top (I_d - B_F B_F^\top) B_H = V^\top B_{\text{res}}.$$

Let $B_{\text{res}} = Q \Sigma R^\top$ be a full singular value decomposition, where $Q = [q_1, \dots, q_d]$ and $\sigma_1 \geq \dots \geq \sigma_d \geq 0$. Then

$$\|V^\top B_H\|_F^2 = \|V^\top B_{\text{res}}\|_F^2 = \sum_{i=1}^d \sigma_i^2 \|V^\top q_i\|_2^2.$$

Set $a_i = \|V^\top q_i\|_2^2$. Since $V^\top V = I_k$ and Q is orthogonal, we have $0 \leq a_i < 1$, $\sum_{i=1}^d a_i = \text{tr}(V^\top Q Q V) = k$. Because the singular values are ordered decreasingly, the weighted sum is maximized by assigning unit weight to its first k terms:

$$\sum_{i=1}^d \sigma_i^2 a_i \leq \sum_{i=1}^k \sigma_i^2.$$

Choosing $V = U = [q_1, \dots, q_k]$ attains this bound. Moreover, $U^\top B_H = U^\top B_{\text{res}}$, and therefore

$$\|V^\top B_H\|_F^2 \leq \|U^\top B_H\|_F^2 = \sum_{i=1}^k \sigma_i^2(B_{\text{res}}).$$

This completes the proof. $\square$

A.2 PROOF OF PROPOSITION 2

Let $Q_H = (T_\Pi)_\# \tilde{P}_H$ and $Q_F = (T_\Pi)_\# \tilde{P}_F$, where $T_\Pi(z) = \Pi z$. Consider any coupling $\gamma \in \Pi(Q_\Pi, \tilde{P}_F)$ and let $(X, Y) \sim \gamma$. The decomposition into the retained and removed subspaces gives

$$\begin{aligned} \|X - Y\|_2^2 &= \|\Pi(X - Y)\|_2^2 + \|(I_k - \Pi)(X - Y)\|_2^2 \\ &= \|\Pi(X - Y)\|_2^2 + \|(I_k - \Pi)Y\|_2^2. \end{aligned}$$

The joint distribution of $(X, \Pi Y)$ is a coupling between Q_H and Q_F . Therefore,

$$\mathbb{E}_\gamma [\|X - \Pi Y\|_2^2] \geq W_2^2(Q_H, Q_F).$$

The second term depends only on the marginal distribution of Y , thus

$$\mathbb{E}_\gamma [\|(I_k - \Pi)Y\|_2^2] = \mathbb{E}_{Y \sim \tilde{P}_F} [\|(I_k - \Pi)Y\|_2^2].$$

It follows that

$$W_2^2(Q_H, \tilde{P}_F) \geq W_2^2(Q_H, Q_F) + \mathbb{E}_{Y \sim \tilde{P}_F} [\|(I_k - \Pi)Y\|_2^2].$$

To obtain the reverse inequality, let η be an optimal coupling between Q_H and Q_F . For each projected point u , sample Y according to the conditional distribution of $\tilde{P}_F$ given $\Pi Y = u$. This lifts η to a coupling between Q_H and $\tilde{P}_F$ for which $(X, \Pi Y) \sim \eta$. Under this coupling,

$$\begin{aligned} \mathbb{E} [\|X - Y\|_2^2] &= \mathbb{E} [\|X - \Pi Y\|_2^2] + \mathbb{E} [\|(I_k - \Pi)Y\|_2^2] \\ &= W_2^2(Q_H, Q_F) + \mathbb{E}_{Y \sim \tilde{P}_F} [\|(I_k - \Pi)Y\|_2^2]. \end{aligned}$$

Combining the two inequalities gives

$$W_2^2(Q_H, \tilde{P}_F) = W_2^2(Q_H, Q_F) + \mathbb{E}_{Y \sim \tilde{P}_F} [\|(I_k - \Pi)Y\|_2^2].$$

Let μ_F and Σ_F denote the mean and covariance of $\tilde{P}_F$. Then,

$$\begin{aligned} \mathbb{E} [\|(I_k - \Pi)Y\|_2^2] &= \|(I_k - \Pi)\mu_F\|_2^2 + \text{tr}((I_k - \Pi)\Sigma_F(I_k - \Pi)) \\ &= \|(I_k - \Pi)\mu_F\|_2^2 + \text{tr}((I_k - \Pi)\Sigma_F). \end{aligned}$$

Hence,

$$W_2^2((T_\Pi)_\# \tilde{P}_H, \tilde{P}_F) = W_2^2((T_\Pi)_\# \tilde{P}_H, (T_\Pi)_\# \tilde{P}_F) + \|(I_k - \Pi)\mu_F\|_2^2 + \text{tr}((I_k - \Pi)\Sigma_F).$$

Thus, if the removed subspace contains nonzero faithful energy, the projected distribution remains at a strictly positive Wasserstein distance from $\tilde{P}_F$. This completes the proof. $\square$

A.3 PROOF OF PROPOSITION 3

Let $X = Z - \mu_H$ and $\delta = \mu_F - \mu_H$. Then $\mathbb{E}[X] = 0$, $\text{Cov}(X) = \Sigma_H$, and

$$T_A(Z) - Z = \delta + (A - I_k)X.$$

Therefore,

$$\begin{aligned} \mathbb{E} [\|T^*(Z) - Z\|_2^2] &= \|\delta\|_2^2 + \mathbb{E} [\|(A - I_k)X\|_2^2] \\ &= \|\delta\|_2^2 + \text{tr}[(A - I_k)\Sigma_H(A - I_k)^\top] \\ &= \|\mu_F - \mu_H\|_2^2 + \text{tr}(\Sigma_H) + \text{tr}(A\Sigma_H A^\top) - 2\text{tr}(A\Sigma_H) \\ &= \|\mu_F - \mu_H\|_2^2 + \text{tr}(\Sigma_H) + \text{tr}(\Sigma_F) - 2\text{tr}(A\Sigma_H). \end{aligned}$$

The first three terms are independent of A . Thus, minimizing the expected distortion is equivalent to solving

$$\max_{A: A\Sigma_H A^\top = \Sigma_F} \text{tr}(A\Sigma_H).$$

For any feasible A , define $Q = \Sigma_F^{-\frac{1}{2}} A \Sigma_H^{\frac{1}{2}}$. The feasibility constraint implies

$$QQ^\top = \Sigma_F^{-\frac{1}{2}} A \Sigma_H A^\top \Sigma_F^{-\frac{1}{2}} = I_k.$$

So Q is orthogonal. Moreover, $A = \Sigma_F^{-\frac{1}{2}} Q \Sigma_H^{-\frac{1}{2}}$ and thus,

$$\text{tr}(A \Sigma_H) = \text{tr}(Q \Sigma_H^{-\frac{1}{2}} \Sigma_F^{\frac{1}{2}}).$$

Let $M = \Sigma_H^{-\frac{1}{2}} \Sigma_F^{\frac{1}{2}}$. By the von Neumann trace inequality, for any orthogonal Q ,

$$\text{tr}(QM) \leq \|M\|_* = \text{tr}\left((MM^\top)^{\frac{1}{2}}\right).$$

Since $MM^\top = \Sigma_H^{-\frac{1}{2}} \Sigma_F \Sigma_H^{-\frac{1}{2}}$, we have

$$\text{tr}(A \Sigma_H) \leq \text{tr}\left[(\Sigma_H^{-\frac{1}{2}} \Sigma_F \Sigma_H^{-\frac{1}{2}})^{\frac{1}{2}}\right].$$

We define $S = (\Sigma_H^{-\frac{1}{2}} \Sigma_F \Sigma_H^{-\frac{1}{2}})^{\frac{1}{2}}$. The proposed matrix $A^* = \Sigma_H^{-\frac{1}{2}} S \Sigma_H^{-\frac{1}{2}}$ is feasible. Furthermore,

$$\text{tr}(A^* \Sigma_H) = \text{tr}(\Sigma_H^{-\frac{1}{2}} S \Sigma_H^{-\frac{1}{2}}) = \text{tr}(S),$$

Therefore, A^* attains the upper bound and therefore minimizes the expected distortion. To establish uniqueness, let $M = U \Lambda V^\top$ be the singular value decomposition of M . Because Σ_H and Σ_F are positive definite, all singular values in Λ are strictly positive. For any orthogonal Q ,

$$\text{tr}(QM) = \text{tr}(V^\top Q U \Lambda) \leq \text{tr}(\Lambda).$$

Equality holds only when $V^\top Q U = I_k$. Therefore, the corresponding feasible A is unique. Since A^* attains this maximum, it is the unique minimizer. Consequently, T^* achieves the target first and second moments without requiring Z to be Gaussian. This completes the proof. $\square$

B IMPLEMENTATION AND EVALUATION DETAILS

Backbones and Paired Calibration Data. We evaluate ResOT on LLaVA-1.5-7B, InstructBLIP-Vicuna-7B, and mPLUG-Owl2-LLaMA2-7B. All models are evaluated with greedy decoding to avoid randomness in generation. For each backbone, we use the same 5000 calibration pairs from the training data released with LURE (Zhou et al., 2024). Each pair associates an image from the COCO 2014 training set with a faithful caption from LLaVA-Instruct-150K and a corresponding hallucinated caption generated by GPT-3.5. We separately feed the faithful and hallucinated image-caption inputs through the frozen LVLM and extract their final-position hidden states at the selected decoder layer. During inference, ResOT repairs the hidden state used to predict the next token at each decoding step.

Evaluation Protocols. For POPE, we evaluate on the random, popular, and adversarial splits, each containing 3000 samples. A prediction is parsed by checking whether the first generated word is “yes” or “no”, otherwise it is counted as unknown. For CHAIR, we evaluate object hallucination on the COCO 2014 validation images (Lin et al., 2014). Following prior work (Huang et al., 2024; Yang et al., 2025a), we use the subset of 500 images selected by OPERA for efficient evaluation. The captioning prompt is fixed as: *Please describe this image in detail.* For MME, we use the official yes/no evaluation setting. Each image is associated with two questions, and the final score combines question-level accuracy and image-level pair accuracy. For GPT-assisted open-ended evaluation, we use GPT-4o as an external judge to assess image description quality. Given an image, the question, and the model responses, GPT is asked to score both responses from 1 to 10 in terms of accuracy and detailedness.

Baseline Implementation. We compare with decoding-based methods, including OPERA, VCD, HALC, and LVLMs-Saliency, and representation-level methods, including ICT, VTI, VISTA, Nullu, HulluEdit and REVIS. For each baseline, we use the official code and the hyperparameters recommended in the corresponding paper. All methods use greedy decoding except OPERA, for which we follow the official beam search setting with five beams. Within each benchmark, we keep the LVLM backbone, prompt, and maximum generation length the same across all methods.

Table 5: AMBER results across three LVLM backbones. Lower CHAIR, Hal, and Cog indicate less hallucination, while higher Cover indicates better object coverage.

Backbone	Method	CHAIR ↓	Cover ↑	Hal ↓	Cog ↓
LLaVA-1.5	Baseline	7.6	48.7	35.4	4.2
	Nullu	7.6	49.2	37.8	3.7
	ResOT	6.0	47.8	25.1	2.4
mPLUG-Owl2	Baseline	9.3	50.3	40.8	5.3
	Nullu	6.6	48.3	27.7	3.0
	ResOT	4.6	48.4	18.7	1.6
InstructBLIP	Baseline	7.5	53.8	36.3	3.8
	Nullu	8.6	51.3	33.9	3.6
	ResOT	6.4	52.9	31.1	3.2

Table 6: MMHal-Bench results across three LVLM backbones. Avg. denotes average score, and Hall. denotes hallucination rate.

Method	LLaVA-1.5		mPLUG-Owl2		InstructBLIP	
	Avg. ↑	Hall. ↓	Avg. ↑	Hall. ↓	Avg. ↑	Hall. ↓
Baseline	2.1250	0.6458	1.9479	0.6667	1.8229	0.6458
Nullu	2.1146	0.6562	1.9583	0.6458	1.6250	0.6979
ResOT	2.2708	0.5938	2.0104	0.6458	1.9583	0.6146

Hyperparameter Settings. ResOT uses two rank parameters. The rank c determines the number of dominant directions used to estimate the faithful and hallucinated subspaces, while k determines the dimension of the residual repair subspace. We set $c = k$ and select this shared rank from $\{8, 16, 32, 64, 128\}$. The repair strength α is selected from $\{0.1, 0.2, \dots, 1.0\}$. At each decoding step, repair is applied only to the final-position hidden state in decoder layers 16-31. We set the covariance shrinkage coefficient to $\rho = 0.05$ and the numerical stability term to $\epsilon = 10^{-6}$. For each layer, the correction threshold is set to the 95th percentile of OT correction norms computed from the hallucinated calibration representations. These settings are used across all models and benchmarks.

C ADDITIONAL EXPERIMENTAL RESULTS

C.1 RESULTS ON ADDITIONAL BENCHMARKS

We further evaluate ResOT on AMBER, MMHal-Bench, and MM-Vet (Wang et al., 2023; Sun et al., 2024; Yu et al., 2023). AMBER provides fine-grained analysis of object hallucination, MMHal-Bench evaluates hallucination in open-ended responses, and MM-Vet measures general multimodal capability across six skills. For AMBER, we use its generative split with the official prompt *Describe this image.* and report CHAIR, Cover, Hal, and Cog using the official LLM-free evaluator. For MMHal-Bench, we answer all 96 questions and use GPT-4o with the official evaluation prompt to report the average score and hallucination rate. For MM-Vet, we answer all 218 questions using their original prompts and follow the official GPT-assisted evaluation protocol.

Results on AMBER. Table 5 shows that ResOT consistently reduces hallucination on AMBER. It achieves the lowest CHAIR, Hal, and Cog values across all three backbones, demonstrating fewer hallucinated object mentions, fewer responses containing hallucinations, and less hallucination caused by contextual associations. These gains are achieved while preserving comparable object coverage, with Cover remaining within 1.9 points of the best result for each backbone.

Results on MMHal-Bench. Table 6 shows that ResOT achieves the highest average score across all three backbones while reducing hallucination over the corresponding baselines. It obtains the lowest hallucination rate on LLaVA-1.5 and InstructBLIP and ties the best result on mPLUG-Owl2, demonstrating improved open-ended response quality and faithfulness.

Table 7: MM-Vet results across three LVLM backbones. We report scores on recognition (Rec.), OCR, knowledge (Know.), generation (Gen.), spatial reasoning (Spat.), math, and the overall total score. Higher scores indicate better performance.

Backbone	Method	Rec.	OCR	Know.	Gen.	Spat.	Math	Total
LLaVA	Vanilla	34.0	22.6	16.1	20.1	24.4	11.5	29.5
	Ours	34.6	23.2	16.3	19.4	27.1	11.5	31.0
mPLUG-Owl2	Vanilla	38.7	27.0	23.8	25.4	29.5	3.8	34.3
	Ours	38.2	28.2	23.6	24.9	30.9	3.8	34.5
InstructBLIP	Vanilla	30.9	15.9	15.4	16.0	17.3	8.1	25.8
	Ours	33.5	17.1	19.6	18.0	18.1	7.3	27.7

Table 8: Inference efficiency on LLaVA-1.5 with 128 generated tokens. Changes are computed against each method’s paired Vanilla run.

Method	Tokens/s (vs. Vanilla) $\uparrow$	Peak Memory Increase (MiB) $\downarrow$	Preserves LVLM Weights
Vanilla	49.326 (0.00%)	0.000	✓
OPERA	0.752 (−97.87%)	25219.631	✓
VCD	17.462 (−52.64%)	724.160	✓
HALC	1.858 (−95.21%)	4728.521	✓
ICT	24.285 (−39.46%)	0.250	✓
VTI	29.940 (−24.77%)	56.504	✓
VISTA	22.200 (−51.43%)	3761.718	✓
Nullu	45.850 (−1.64%)	0.000	✗
HulluEdit	17.843 (−56.16%)	112.595	✓
ResOT	40.071 (−18.76%)	8.147	✓

Results on MM-Vet. Table 7 shows that ResOT improves the overall MM-Vet score across all three backbones, with gains spanning multiple multimodal skills. Together with the hallucination results, these findings demonstrate that ResOT improves visual faithfulness while maintaining or enhancing general multimodal capability.

C.2 COMPLEXITY ANALYSIS

ResOT estimates the residual bases and Gaussian transport map offline, while online inference only applies the transport maps to the current token representations. Let L denote the number of repaired layers. At each layer, projecting the hidden state into the residual subspace and reconstructing it in the original hidden space require two matrix vector multiplications, each with a cost of $O(dk)$. Applying the affine transport map costs $O(k^2)$, while correction clipping costs $O(k)$. The likelihood gate is computed once for each generated token at the first layer and is shared by all repaired layers. Computing the gate costs $O(k^2)$. Therefore, the total additional complexity for each generated token is $O(M(dk + k^2) + k^2)$. Since $k \ll d$, the dominant term is $O(Mdk)$. The persistent repair state requires $O(M(dk + k^2))$ memory and introduces no trainable parameters.

As shown in Table 8, ResOT retains 81.24% of the Vanilla throughput and increases only 8.147 MiB of peak GPU memory. It also keeps the LVLM backbone weights unchanged. ResOT therefore has low deployment overhead and can be enabled or disabled without changing the model checkpoint. Nullu is faster, but it directly modifies the backbone weights. Returning to the original model requires restoring these weights. Its suppression operation may also remove useful representations and affect general multimodal performance.

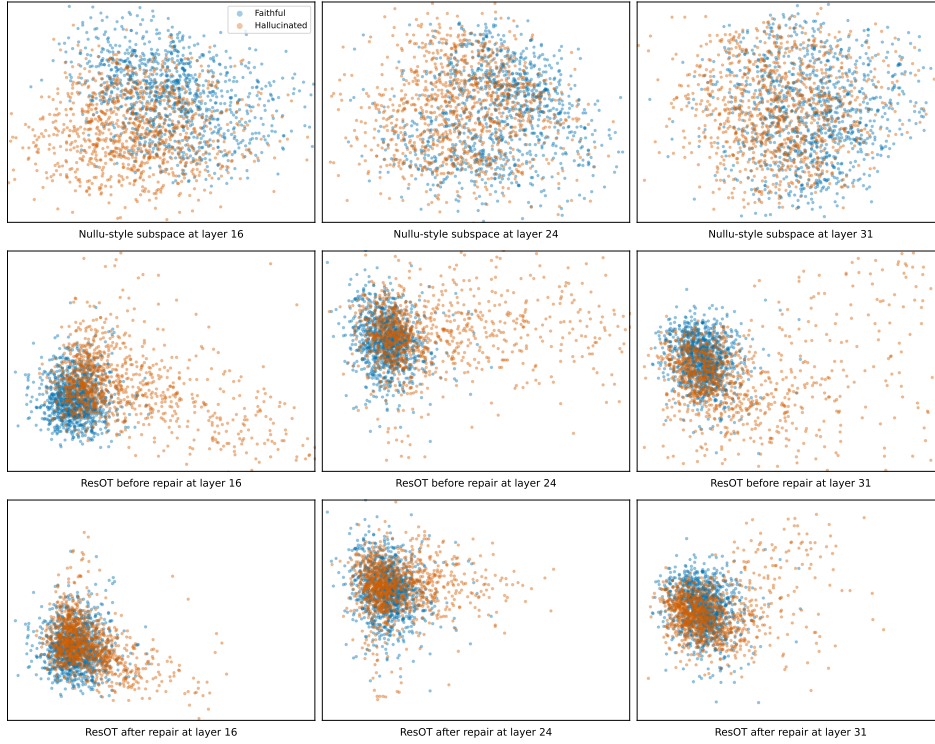


Figure 6: Multi-layer representation visualization at layers 16, 24, and 31. Rows correspond to the Null-style subspace, the ResOT residual subspace before repair and after repair.

Table 9: Separability between faithful and hallucinated LLaVA representations in the subspace on the evaluation split. Higher values indicate stronger separability.

Layer	Setting	Mahalanobis Dist. $\uparrow$	LDA Acc. $\uparrow$	LDA AUC $\uparrow$
16	Null-style	1.5033	0.7795	0.8674
	ResOT before repair	2.4725	0.8865	0.9544
24	Null-style	1.3405	0.7590	0.8366
	ResOT before repair	2.1708	0.8565	0.9351
31	Null-style	1.2530	0.7325	0.8161
	ResOT before repair	1.9297	0.8460	0.9149

D SENSITIVITY AND ABLATION ANALYSIS

D.1 QUANTITATIVE ANALYSIS OF REPRESENTATION

For all analyzes in this subsection, we use the first 4000 faithful-hallucinated pairs for fitting and reserve the remaining 1000 pairs for evaluation. All visualizations and quantitative results are computed on the held-out split.

Visualization across Layers. Figure 6 extends the visualization in Figure 4 to layers 16 and 24. Across all three layers, the residual coordinates reveal a clearer mismatch between faithful and hallucinated representations than the space after Nullu suppression. This consistent pattern shows that filtering out the dominant faithful structure concentrates hallucination-related deviations in the localized repair space. Gaussian OT then aligns the hallucinated residual distribution with the faithful distribution at each layer, confirming that the localization and repair pattern observed in the main text also holds at the other selected layers.

Table 10: Center-distance analysis across layers. F , H , and R denote faithful, hallucinated, and repaired representations, respectively.

Layer	Nullu ($F \rightarrow H$)	Before repair ($F \rightarrow H$)	After repair ($F \rightarrow R$)
16	1.9758	1.2845	0.0537
24	4.0293	2.9075	0.1265
31	4.4187	2.5757	0.0994

Table 11: Average relative displacement of faithful and hallucinated samples in the representation space. H/F denotes the ratio of hallucinated to faithful displacement. The 16–31 row reports the average across all selected layers.

Layer	Faithful (%)	Hallucinated (%)	H/F
16	1.5903	7.1353	4.48×
24	1.1953	5.9745	5.00×
31	1.3840	4.2196	3.05×
16–31	1.3425	6.0281	4.49×

Subspace Separability. To quantify the localization achieved by subspace construction, we compare the separability of faithful and hallucinated representations in the space after Nullu suppression and the ResOT residual space before repair. As shown in Table 9, the residual space consistently yields higher Mahalanobis distance, LDA accuracy, and LDA AUC across all evaluated layers, demonstrating that it better isolates hallucination-related deviations.

Center Alignment by Gaussian OT. Table 10 further quantifies the repair effect. Faithful and hallucinated centers are clearly separated in the residual subspace before repair. After applying Gaussian OT, the distance between the faithful and repaired centers converges to near zero across all evaluated layers. This demonstrates that Gaussian OT closes the localized center mismatch through transport rather than hard suppression.

Selectivity of Representation Repair. Table 11 shows that hallucinated representations consistently receive larger updates than faithful representations across layers. This pattern reflects the overall inference mechanism of ResOT, which combines OT correction within the residual subspace with adaptive control of repair strength. This demonstrates that ResOT concentrates changes on representations with stronger hallucination evidence rather than shifting all representations uniformly.

D.2 HYPERPARAMETER SENSITIVITY

Figure 7 analyzes the sensitivity of ResOT to the repair strength α and residual subspace rank on LLaVA-1.5. With the rank fixed at 32, increasing α generally lowers C_s and improves BLEU, while MME peaks at $\alpha = 0.8$ and drops beyond this point. Smaller ranks favor C_s and BLEU, whereas ranks above 32 weaken all three metrics. Overall, rank 32 and $\alpha \in [0.7, 0.8]$ provide a favorable balance among hallucination reduction, captioning quality, and general multimodal capability.

D.3 GAUSSIAN DIAGNOSTICS IN THE HALLUCINATION RESIDUAL SUBSPACE

Q1: Are the residual coordinates approximately Gaussian? Figure 8 shows the diagnostics at layer 31. We first project faithful and hallucinated hidden states into the hallucination residual subspace. We then whiten each distribution using its estimated mean and covariance. The marginal histograms largely overlap with the standard normal reference. The marginal Q-Q plots also stay close to the diagonal in the central quantiles. Figure 9 shows a similar pattern across layers 16–31. The marginal KS distances are especially small for faithful representations. These results show that the Gaussian approximation describes the central region of individual coordinates reasonably well.

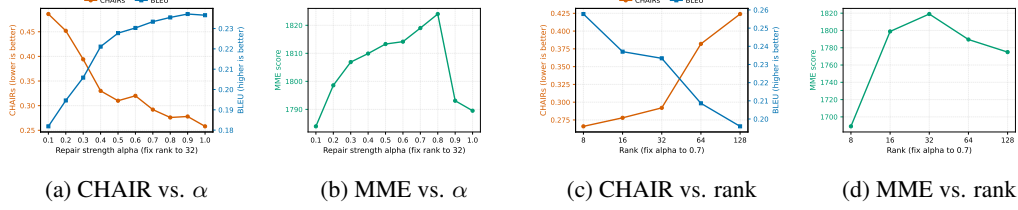


Figure 7: Sensitivity analysis of the repair strength α and residual subspace rank on LLaVA-1.5. For α sensitivity, we fix the rank to 32. For rank sensitivity, we fix α to 0.7.

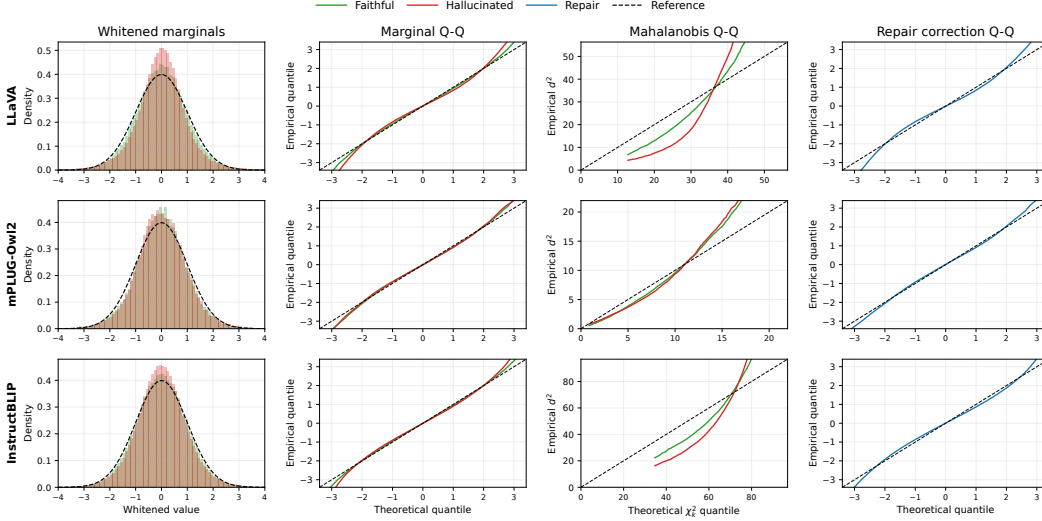


Figure 8: Gaussian diagnostics at layer 31 for LLaVA, mPLUG-Owl2 and InstructBLIP. The columns show whiten marginals, marginal Q-Q plots, Mahalanobis Q-Q plots and whiten repair corrections. The correction is $T^*(z_h) - z_h$. Its norm is clipped using the 95th percentile of the calibration correction norms. This is done before multiplication by the repair strength and the likelihood gate.

Q2: Does marginal normality imply that the joint distribution is Gaussian? No. The Mahalanobis Q-Q plots depart more clearly from the χ_k^2 reference. The corresponding KS distances are also larger. These differences are most visible in the tails. A coordinate can appear nearly Gaussian on its own even when the full distribution is not multivariate Gaussian. The normalized mean squared Mahalanobis distance, $\mathbb{E}[d^2]/k$, gives an additional check of the overall scale. However, a value near one does not prove joint Gaussianity. The Gaussian model is therefore a useful approximation of the first two moments, but it is not an exact model of the full distribution.

Q3: Does the diagnostic examine the correction that is actually applied? Yes. Rather than introducing an unrelated residual, we analyze the repair correction produced by the estimated OT map. For each hallucinated coordinate z_h , we compute

$$\delta_{\text{raw}} = T^*(z_h) - z_h, \quad \delta_{\text{clip}} = \min\left(1, \frac{\tau_{0.95}}{\|\delta_{\text{raw}}\|_2 + \epsilon}\right) \delta_{\text{raw}}, \quad (11)$$

where $\tau_{0.95}$ is the 95th percentile of the correction norms from the same 5,000 calibration pairs used to fit the OT map. The final column of Figure 8 shows the whiten δ_{clip} . At this stage, it has not been multiplied by the repair strength α or the likelihood gate. Its central quantiles are approximately linear, while its tails depart from the Gaussian reference. The normalized mean squared Mahalanobis distance can fall below one. This occurs because norm clipping limits large correction norms and because the covariance used for whitening is regularized.

Q4: Why is Gaussian OT still appropriate, and how are tail deviations handled? Our method uses the Gaussian model to estimate transport from the mean and covariance. It does not assume

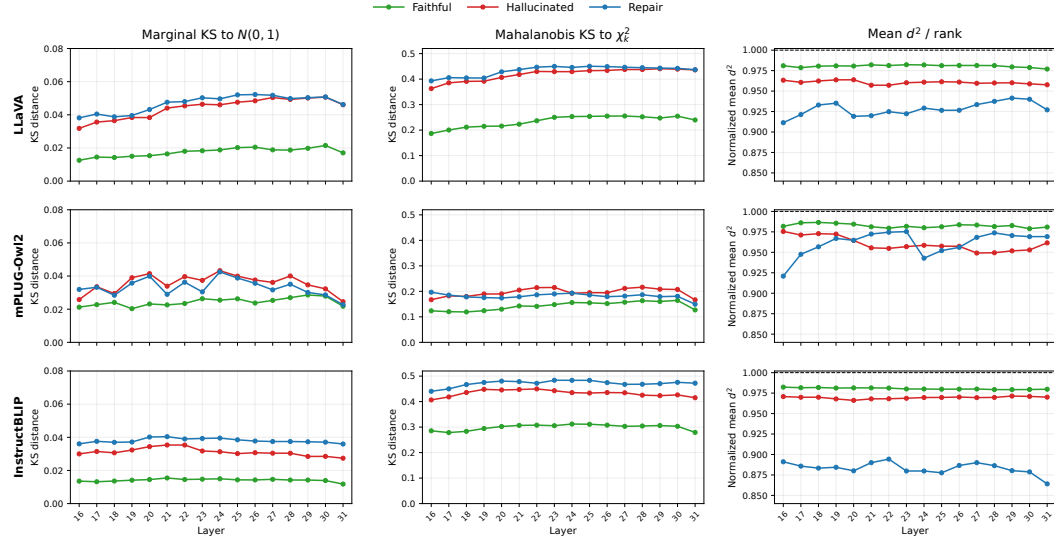


Figure 9: Gaussian diagnostics across layers 16–31 for LLaVA, mPLUG-Owl2 and InstructBLIP, using the same 5,000 calibration pairs as the OT maps. The columns report the marginal KS distance to $N(0, 1)$, the Mahalanobis KS distance to χ_k^2 and $\mathbb{E}[d^2]/k$. The curves compare faithful coordinates, hallucinated coordinates and repair corrections after norm clipping.

that every projected activation is exactly Gaussian. The diagnostics show that this approximation captures the main structure in the central region. They also reveal deviations in the joint tails. We use covariance shrinkage to stabilize the estimated transport map. We then clip correction norms at the 95th percentile to bound unusually large interventions. These measures retain the simple Gaussian OT solution while limiting the influence of the observed tail deviations.

E LIMITATION

ResOT intentionally uses a low dimensional Gaussian approximation to obtain a closed-form transport map. This design captures the main distribution shift through mean and covariance, rather than modeling every detail of the residual distribution. A useful direction is to explore richer transport families while retaining the current inference efficiency. In addition, our evaluation mainly focuses on object hallucination. Evaluating ResOT on other types of multimodal hallucination would provide a more complete understanding of its applicability.